Summative Assessment 2

Machine Learning Models for Synthesizing and Forecasting the Electricity Outage Risk Index using Climate Time-Series Data

Far Eastern University - Manila

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Abstract

The Philippines, situated within the Pacific typhoon belt, is highly susceptible to frequent and intense weather phenomena, particularly destructive typhoons. This geographical reality poses a significant and recurrent threat to the nation's critical energy infrastructure, compromising the reliability of electricity access for millions. Proactive and accurate prediction of weather-related grid instability is paramount for effective disaster preparedness and resource pre-positioning. Developing such systems is, however, severely constrained by a fundamental data deficiency, the lack of publicly accessible, high-resolution, and consistent historical electricity outage data, which prevents the direct use of a true ground-truth dependent variable.

This study addresses this challenge by pioneering a Machine Learning regression framework to predict a synthesized Electricity Outage Risk Index. Ranging from a score of 1 to 100, the EORI functions as a quantifiable proxy for grid vulnerability, constructed from a meticulous synthesis of key climate and atmospheric indicators known to critically influence grid integrity. Through systematic comparative analysis, Least Angle Regression was selected for modeling system loss, and LASSO regression proved optimal for forecasting power consumption. This selection offers a robust and highly interpretable approach, leveraging the regularization properties of both techniques for feature selection and model stability.

A central feature of the proposed system is the innovative user-configurable weighting mechanism. This acknowledges the inherent uncertainty and regional variability in defining a monolithic risk index. To maximize localized utility, stakeholders are provided with 100 allocation units to distribute among the core climatic features comprising the EORI. This unique functionality allows for the customization of the risk index based on region-specific infrastructure knowledge or evolving perceived risk priorities. Ultimately, the resulting framework delivers a robust, interpretable, and highly adaptable tool, significantly enhancing the capability of energy providers and policymakers to strategically mitigate weather-related energy risks and bolster national energy security.

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List of Acronyms

- **EORI** – Electricity Outage Risk Index
- **FOI** – Freedom of Information
- **IDE -** Integrated Development Environment
- **LARS** - Least Angle Regression
- **LASSO** - Least Absolute Shrinkage and Selection Operator
- **MAE** – Mean Absolute Error
- **MCDA** – Multi-Criteria Decision Analysis
- **MADM** – Multi-Attribute Decision Making
- **ML** – Machine Learning
- **OECD** – Organisation for Economic Co-operation and Development
- **SHAP** – SHapley Additive exPlanations
- **XGBoost** – Extreme Gradient Boost

Chapter 1: Introduction

1.1 Project Overview and Context

The provision of a stable and reliable electricity supply is a fundamental requirement for sustained national development, economic growth, and public safety. As a developing country, the Philippines faces the persistent challenge of balancing rapid industrial and demographic expansion with the capacity and resilience of its existing energy infrastructure. The national power sector still relies heavily on centralized, often aging, fossil fuel sources, and the transmission and distribution network frequently lacks the necessary redundancy to ensure continuous service. This infrastructural fragility means that maintaining power stability is a constant vulnerability, making the system highly susceptible to even moderate external shocks.

This vulnerability is first critically exposed by the country's placement just above the equator. This location contributes to consistently high temperatures, magnifying the effects of global warming. The resulting extreme heat puts severe stress on power generation assets, forcing thermal power plants to operate at reduced efficiency and capacity (Philippine Disaster Resilience Foundation, 2024). Operationally, this leads to the declaration of grid warnings, such as yellow or red alerts, and in extreme cases, necessitates forced outages to prevent damage or cascading system failures (Crismundo, 2025; Baclig, 2024). This sustained heat-related stress compromises the grid's operational margin and reduces its ability to handle peak demand.

Further compounding the risk is the Philippines' location along the Pacific typhoon belt, which makes it a primary target for intense tropical cyclones. The country experiences an exceptional frequency of these events, with an average of 20 typhoons annually. The worsening intensity of these storms, driven by climate change, generates powerful winds, storm surges, and catastrophic flooding that down several power lines, cause substantial structural damage to transmission towers, and disrupt substation operations (Cabuenas, 2025; Esmael, 2025). This highlights the lack of typhoon-resilient energy facilities in the country that can withstand the worsening typhoons that hit the country each year (Villanueva, 2025). These weather elements, specifically typhoon count, precipitation, and wind speed, are directly responsible for most physical damage and subsequent prolonged outages.

Given the systemic risks posed by vulnerable infrastructure combined with the intensifying frequency of climate shocks, the traditional methodology of reactive post-disaster recovery is proving unsustainable. A critical and systematic shift toward proactive risk prediction and mitigation is required to safeguard economic stability and public welfare. This research addresses this need by utilizing Machine Learning to develop a quantifiable, forward-looking tool. The resulting model will be trained not only on direct weather factors but also on supplementary climate indicators to forecast the Electricity Outage Risk Index, a predicted score from 1 to 100. A novel feature of the system allows end-users to apply 100 credits to customize the weights of these variables, ensuring the tool's predictions are highly relevant to specific local operational contexts.

1.2 Problem Statement

The intensifying threat of climate change to the national power grid is compounded by a critical data deficiency, severely impeding the development of effective, proactive solutions. The fundamental challenge of this research lies in the absence of publicly accessible, granular, historical data detailing the monthly frequency and magnitude of electricity outages across the Philippines (NewsBytes.ph, 2025). This data gap is the primary constraint that prevents the application of advanced, conventional supervised Machine Learning models, which rely on a direct, ground-truth dependent variable (Barro, 2025).

This scarcity is severe despite some success in acquiring public records. For example, a Freedom of Information request submitted on September 20, 2023, resulted in the release of aggregate government datasets by October 6, 2023 (Freedom of Information, 2023). While this demonstrates some level of cooperation, the returned information, such as monthly system-wide power consumption and losses, lacks the necessary granularity and specificity required to correlate weather events with specific infrastructure directly. Furthermore, the detailed outage data that can correlate infrastructure assets with weather events is proprietary, held by private utility companies. Acquiring this level of detail typically requires significant financial investment, creating a prohibitive cost barrier that limits the ability of external stakeholders, researchers, and disaster risk management personnel to develop sophisticated, localized predictive models.

This research addresses this critical data and operational gap directly by establishing a novel and accessible solution. The project will not focus on modeling the historical effects, but instead shifts the predictive focus to the causative metrics, the meteorological and oceanic variables known to trigger grid damage. This approach utilizes readily available, public-domain data to predict an essential proxy metric, the Electricity Outage Risk Index.

The EORI is a synthesized risk score. The core problem statement is, thus, to successfully develop and validate a high-performance regression model that can accurately predict this synthesized EORI, providing a much-needed quantifiable risk forecast that empowers external stakeholders and researchers to conduct further analysis and aids the transition to a proactive, data-driven strategy for Philippine energy resilience planning.

1.3 Project Goal & Objectives

The Project Goal is to develop a flexible and data-driven Machine Learning regression framework that accurately predicts a customized Electricity Outage Risk Index, scaled from 1 to 100, for the Philippines. This initiative aims to transition energy sector stakeholders from a reactive recovery posture to one of proactive resource management, thereby fundamentally enhancing the resilience of the national power grid against recurrent weather-related disruptions.

The concrete, measurable steps required to achieve this goal are listed below:

- To source, clean, and preprocess time-series publicly available weather data, and to mathematically construct an innovative and synthesized Electricity Outage Risk Index through the Data Preparation and EORI Construction phase. This index must be scaled from 1 to 100 and based on normalized features, including typhoon count, precipitation, and wind speed, with completion set for the initial phase of the methodology.
- To develop, tune, and rigorously compare a suite of candidate regression models using key weather features. The final model selection, LARS for system loss and LASSO for power consumption, must achieve a high predictive performance as measured by a Mean Absolute Error of 5 points or less on the 1-100 EORI test set scale through the Model Development and

Optimization phase. This metric must be established and confirmed before the final evaluation stage.

- To design and implement a user-facing mechanism, employing a functional prototype interface that allows end-users to dynamically adjust the weights of the EORI's underlying components using a budget of 100 allocation units through the System Flexibility and Customization phase. The final system must be functionally demonstrated and deployed as part of the final system deployment.
- To empirically evaluate the primary LARS and LASSO models against simpler baselines and thoroughly analyze the model's structure using SHapley Additive exPlanations through the Performance and Feature Analysis phase. This analysis must quantify the relative contribution of each climate factor and be fully detailed within the Results and Discussion chapter.

1.4 Proposed Solution

The proposed solution is the development of a Machine Learning-based predictive framework centered on carefully selected regression models. This research approach directly addresses the core problem of data scarcity by shifting the predictive focus from the unobserved actual outage data to the causative weather metrics.

The solution's core innovation lies in its approach to data deficiency. Due to the unavailability of granular outage records, the system's core function is to forecast a synthesized variable, the Electricity Outage Risk Index (EORI). The model utilizes a range of time-series climatic and oceanic features, including key destructive elements like typhoon count, precipitation, and wind speed, alongside supplementary indicators like oceanic index and wind shear, to model the complex, non-linear relationships that collectively determine grid vulnerability. The EORI provides a continuous, normalized risk score ranging from 1 to 100, offering a simple yet actionable metric for monthly grid vulnerability.

The expected outcome of this research is a highly quantifiable, proactive risk assessment tool that delivers three key advantages for policy and operational planning.

First, the predictive system utilizes robust and interpretable linear models, specifically, Least Angle Regression for predicting system loss and LASSO for power consumption. These models were selected after rigorous comparison for their optimal balance of predictive accuracy, computational efficiency, and transparency.

Second, the system includes a crucial component designed for practical use: a user-customization layer where stakeholders can dynamically adjust the EORI's composition using a budget of 100 allocation credits. This ensures the risk index is adaptable to localized priorities and infrastructure profiles.

Third, and critically, the framework incorporates the SHapley Additive exPlanations methodology. While the selected linear models are inherently interpretable, SHAP analysis provides a unified, model-agnostic framework for quantifying the exact contribution of each climatic variable to the final predicted EORI score. This enhances transparency and provides actionable insights.

This unique combination of tailored model selection, user-driven flexibility, and robust explainability ensures the system's output is highly adaptable to specific local infrastructure vulnerabilities and operational priorities, overcoming the limitation of a rigid risk index.

Chapter 2: Review of Related Literature and Theory

2.1 Theoretical Foundations

The research is fundamentally grounded in three primary theoretical domains: Supervised Machine Learning Regression, Composite Indexing, and Regularization and Sparse Modeling Theory.

The project's core methodology is grounded in Supervised Machine Learning Regression, a predictive modeling framework in which an algorithm learns a functional relationship between input variables and a continuous target by analyzing labeled training data. According to Nasteski (2017), supervised learning relies on known input–output pairs to construct a mapping function capable of generating accurate predictions for unseen data, with regression specifically addressing problems involving continuous outcomes. In this study, regression analysis is applied to model the relationship between climatic variables and the Electricity Outage Risk Index, whose continuous nature makes regression the appropriate methodological choice. This approach is consistent with prior applications in climate and energy forecasting; for example, Hejase and Assi (2012) demonstrated the effectiveness of regression-based time-series models in predicting continuous environmental and energy-related measures. In line with standard supervised learning practices, the performance of the model in this project is evaluated based on its ability to minimize the error between predicted and actual outage-risk values, thereby reflecting the accuracy of the learned mapping function.

Since direct and granular outage data is unavailable, the target variable is constructed using the principles of Composite Indexing and Multi-Criteria Decision Analysis (MCDA). According to the Organisation for Economic Co-operation and Development (2008), composite indicators are necessary when a phenomenon is multidimensional and cannot be directly observed, requiring the aggregation of multiple measurable variables into a single metric. The Electricity Outage Risk Index (EORI) thus serves as a synthesized proxy, combining key weather-based factors to represent underlying grid vulnerability. Its design allows for user-defined weighting, which is theoretically supported by Multi-Attribute Decision Making (MADM) principles that emphasize tailoring weights to context-specific priorities and operational needs. Research in customizable risk assessment further confirms this approach: Waqdan et al. (2024) show that enabling stakeholders to adjust attribute weights enhances the relevance and decision-support value of a risk index. In this way, the EORI balances methodological rigor with contextual adaptability, ensuring that different grid operators or regions can appropriately prioritize climatic drivers in assessing outage risk.

The initial selection of the Gradient Boosting Regressor is grounded in both ensemble learning theory and its demonstrated empirical success in predictive modeling across engineering domains. Boosting sequentially combines weak learners to correct residual errors, making it highly effective for capturing the non-linearities inherent in complex climate and materials data. This approach is strongly validated by Singh et al. (2021), who found Gradient Boosting Machine regression to outperform other methods for wind power forecasting, achieving the highest accuracy and lowest error metrics. Similarly, within materials informatics, ensemble methods such as Extreme Gradient Boosting are recognized for their superior performance in predicting mechanical properties of composites, as noted in the comprehensive review by Kibrete et al. (2023). Thus, XGBoost offers a theoretically sound and practically proven framework for robust and high-dimensional regression, ensuring both predictive power and generalization in climate-related modeling tasks.

However, upon further evaluation, it was found that the methodological focus must be strategically shifted to leverage the principles of Regularization and Sparse Modeling Theory, implementing the Least Absolute Shrinkage and Selection Operator Regression model. This transition prioritizes model interpretability and feature selection by adding an L1 penalty $\lambda \sum_{j=1}^{p} |\beta_j|$ to the standard Ordinary Least Squares objective function, which actively shrinks the coefficients β_j of less relevant predictor variables to zero, thus retaining only the most influential climatic factors driving the EORI (Guo et al., 2024). This sparsity directly enables grid operators and policymakers to identify and quantify the individual, significant drivers of outage risk, thereby enhancing the overall transparency and utility of the index. Furthermore, the Least Angle Regression (LARS) algorithm is employed for computational efficiency, providing a path to generate the entire spectrum of solutions for the Lasso problem and ensuring the optimal, most interpretable sparse model can be efficiently identified (Filips et al., 2020). The combined use of the Lasso objective and the LARS algorithm ensures the final EORI model is both statistically sound and meets the core project requirement of generating a highly interpretable, policy-relevant result.

2.2 Domain-Specific Literature Review

The Philippines is at a critical juncture where rapid demographic growth intersects with high climate exposure, making the vulnerability of its power system a central concern. The country's energy infrastructure faces a dual threat from climate-related stressors: acute physical damage from tropical cyclones and chronic operational strain from extreme heat.

Tropical cyclones are a leading cause of widespread and prolonged power outages in Southeast Asia and island nations, damaging transmission towers, substations, and overhead lines through high winds and flooding (Baldwin et al., 2023; Meregillano & Delina, 2023). In the Philippines, aging infrastructure and limited system redundancy exacerbate this vulnerability, leaving the grid highly susceptible to repeated, high-intensity shocks (Rehak et al., 2022; Robielos et al., 2020). At the same time, rising temperatures driven by climate change reduce the efficiency and capacity of thermal power plants and grid equipment, leading to derating and supply deficits during peak demand periods (Guddanti et al., 2025; Baligad, 2024). This chronic stress, combined with acute cyclone damage, underscores the urgent need for proactive resilience planning that integrates climate forecasting, infrastructure hardening, and adaptive grid management (Rehak et al., 2022; Singh & Hachem-Vermette, 2022; Robielos et al., 2020, Navarro & Camara, 2023).

2.3 Review of Methodological Approaches

The complexity of predicting a future and synthetic risk index, like the EORI, based on future climatic variables necessitates a robust two-stage predictive framework. This cascading approach, where the output of one model serves as the input for the next, is a widely accepted methodology in advanced forecasting, supported by literature across various engineering and climate science domains. The first stage addresses the temporal nature of the input features. Since climatic variables are inherently time-dependent, their prediction requires dedicated time-series models that can account for underlying temporal patterns such as trend and seasonality (Mystakidis et al., 2024). Traditional statistical approaches, specifically the Seasonal Autoregressive Integrated Moving Average model, provide a foundational benchmark for forecasting meteorological quantities, demonstrating the ability to capture complex seasonal dynamics in variables such as air temperature and precipitation (Murat et al., 2018; Dimri et al., 2020). However, the energy and climate forecasting landscapes increasingly favor powerful, non-linear techniques. Comparative studies confirm that data-driven methods, such as those employing Extreme Gradient Boosting, often surpass conventional ARIMA models in various time-series applications, including multi-step prediction strategies (Lv et al., 2021). The project leverages a mix of these proven time-series approaches for Stage 1. This strategic choice is critical, as any error introduced in forecasting the climatic variables is known to be compounded and propagated into the final EORI prediction of the cascading system. The design of the second stage, where the forecasted climatic inputs are synthesized to predict the EORI, is justified by the utility of cascading and stacked predictive architectures. This architecture is necessary for high-stakes modeling where the target variable is a function of predictions rather than observations. For instance, complex microgrid optimization systems, like DeepEMS, successfully utilize a hybrid multi-stage machine learning model to integrate diverse energy sources (Safari et al., 2024). Similarly, in climate science, cascaded diffusion models are employed for tropical cyclone prediction, where the output of atmospheric data models feeds directly into the final trajectory prediction (Nath et al., 2024). The selection of the XGBoost Regressor as an EORI predictor is supported by its success in stacked ensemble frameworks (Ramesh & Kumaresan, 2025).

To allow the final model to be effectively synthesized for the predictions from the prior time-series models, lending improved accuracy and robustness to the final EORI prediction, LASSO and LARS regressions are employed as the meta-learner within a stacked ensemble architecture. This is a crucial step for synthesizing the predictions generated by the Stage 1 models, lending improved accuracy and robustness to the final EORI prediction (Jafar et al., 2023; Li et al., 2023).

SHapley Additive exPlanations or SHAP is a powerful method within the domain of Explainable Artificial Intelligence, also known as XAI, and is designed to enhance the transparency and interpretability of complex machine learning models (Choudhary et al., 2025; van Zyl et al., 2024). Built upon the concept of Shapley values from cooperative game theory, SHAP provides an additive feature attribution, assigning an importance value to each input feature for a specific prediction (Santos et al., 2024). This approach is particularly valuable in applications like feature selection for time series energy forecasting and rolling bearing fault diagnosis, as it can reveal features that degrade prediction accuracy and quantify their individual impact on the final output (van Zyl et al., 2024; Santos et al., 2024). For instance, in water quality index forecasting, integrating SHAP with stacked ensemble models has proven effective in increasing predictive accuracy and providing clear insights into the role of physicochemical parameters (Choudhary et al., 2025).

2.4 Research Gap and Project Contribution

The literature review establishes a critical research and operational gap in climate resilience planning for the Philippine energy sector, driven primarily by challenges in accessing and utilizing reliable operational data for predictive modeling. Current solutions are hampered because electricity network data, particularly for medium- and low-voltage systems, is often non-existent or unavailable globally, presenting a fundamental challenge for power sector resilience and climate change adaptation planning (Arderne et al., 2020). Furthermore, even when outage reports are available, they suffer from critical data quality issues, including inconsistencies, missing values, and unidentified outage causes, which significantly degrade the reliability and utility of standard outage statistics (Žarković et al., 2024). This lack of past, standardized data presents major planning and operation challenges in "first-access" electricity systems (Ustun et al., 2019), forcing resilience planning to remain largely reactive. Thus, the essential gap is the absence of an open-source, data-driven, and forward-looking tool specifically tailored to the national context that can translate widely available, long-term climatic forecasts into an actionable, medium-term risk signal for proactive decision-making.

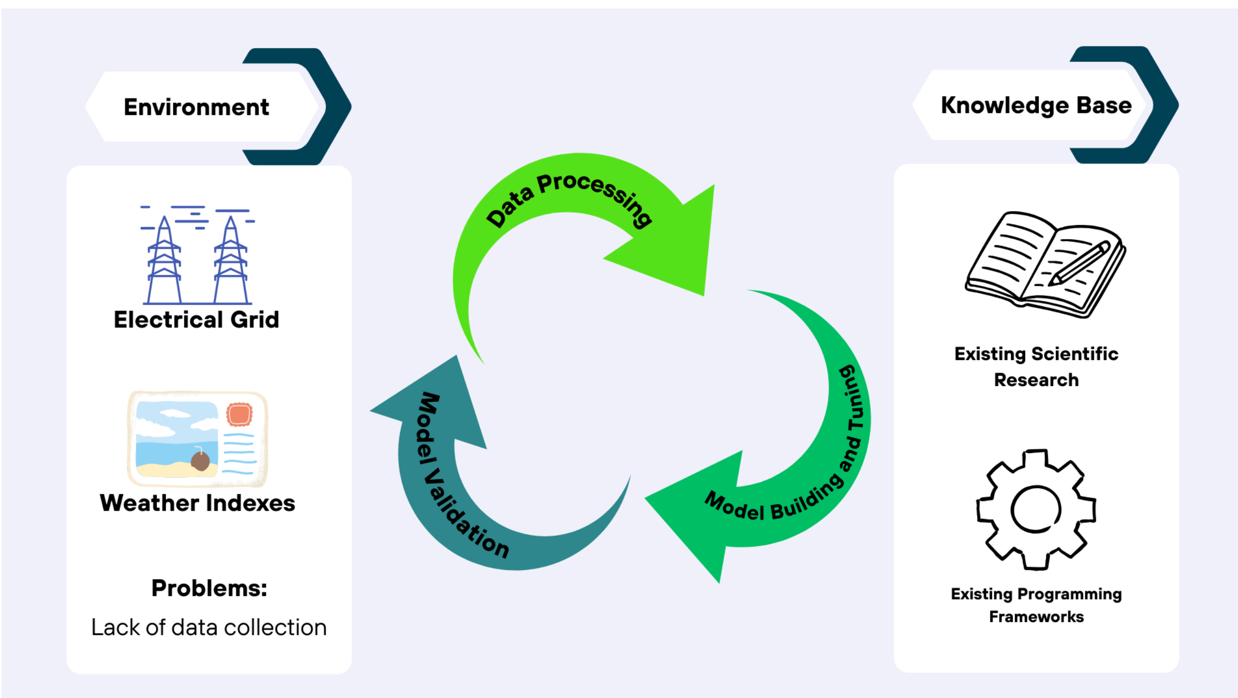
This project directly addresses this deficiency through a novel methodological framework. The central contribution is the development and validation of the Electricity Outage Risk Index. The EORI functions as a synthetic, composite risk metric, a design choice justified by the established practice of using composite risk index methodologies, such as a risk matrix framework, to assess complex policy and climate vulnerabilities in data-constrained environments (Dolge & Blumberga, 2021). This synthetic model approach is further supported by applications in large-scale flood risk mapping and resilience analysis in data-scarce urban and regional areas (Albano et al., 2020; Schinke et al., 2016). By relying entirely on publicly available data, the EORI bypasses the reliance on proprietary outage records. Furthermore, the EORI is integrated into a novel two-stage cascading machine learning architecture. This system explicitly leverages cross-domain time series forecasting to link environmental science with energy security (Smith, 2025). This contribution provides a transparent, predictive resilience tool that facilitates a critical shift from reactive recovery to proactive, risk-informed planning for the future of the Philippine power grid.

Chapter 3: Methodology and Design

3.1 Project Design Overview

In the pursuit of the problem, we have adopted a hybrid experimental research design, by incorporating an Agile development model into a Design Science Research methodology. This methodology is adopted in lieu of the short timeframe of our group to accomplish this product, as this set up will allow us to rapidly encode the data preprocessing, build up the model, and the writing of the paper.

The framework could be seen as follows:



3.2 Data Collection and Preparation

The data related to energy in the Philippine context was sourced from published datasets by the Department of Energy, in a collection entitled Philippine Power Statistics. From here, annual monthly data, from 2020 to 2024, regarding monthly sales, system loss, and total consumption.

On the other hand, weather related data, like the total number of typhoons in a month, or the maximum recorded temperature in a given month, was collected through Meteostat's python api, with the exception of windspeed, which was directly copied from their website.

The combined dataset has 60 data points, 15 feature (weather hazard) columns, and the dependent variables system loss and power consumption (both in MWh units). Missing values were imputed using the median. After cleaning, feature selection of weather hazard features was commenced, using correlational analysis. Multicollinearity is then manually addressed by removing a feature from highly correlating pairs.

After feature selection, a total of 10 weather hazard features were chosen, namely the total number of typhoons in a given month, the mean monthly precipitation total in mm, average wind speed in km/h, maximum and minimum air temperature, both in ◦C, the oceanic nino index, the regional sea surface temperature anomaly near the Philippines, the Madden-Julian oscillation phase, the relative humidity, and the mean sunshine total in minutes.

3.3 Model Selection and Rationale

Given the nature of our problem and the data associated with it, regression models were appropriate for our purposes of interpolating the power consumption during a month, given the data that we have for that given month. For that purpose, two families of regressors come to mind: linear models and ensemble regressors.

Multiple linear models have been picked as candidates: the Least Angle Regression (LARS) model has been picked for its design for modeling high-dimensional data, in which it estimates the coefficients using its covariates, and is thus faster than regular stepwise feature selection (Efron et al., 2004). Linear models with regularization, which add a penalty term to their cost function, are also considered as candidates. These are the Least Absolute Shrinkage and Sum Operator (LASSO) regression, the Ridge regression, and the Elastic Net (which is the combination of the prior two) models. The Lasso Lars regression model, a combination of the LASSO and LARS models, is also considered (Efron et al., 2004). These models, essentially, reduce overfitting by reducing the number of significant features, addressing multicollinearity, and reducing model complexity.

Several ensemble models were also considered due to the robustness of the results owing to their architecture. Bagging ensemble models, which is the Random Forest Regressor in our case, create an ensemble model by first building up smaller weak models made from bootstrapped data, and aggregating their results (IBM, n.d.). This then averages out the variance of the model, therefore reducing overfitting.

Boosting, on the other hand, is not done in parallel, but is instead a step of multiple weak models that are sequentially built, with each step addressing the weaknesses of the last iteration. AdaBoost, Gradient Boosting, and XGBoost are all model candidates in this category.

3.4 Experimental Setup

Before training the model, the dataset is first split into an 80-20 split, with 80 being the training data. In turn, a “Leave One Out” cross-validation (LOOCV) is used. We made this decision due to the sparseness of our dataset, with LOOCV allowing us to account for that said sparseness, while also still conducting cross-validation. Along with this, grid search was used in tuning the hyperparameters.

To investigate the effect of scale, particularly in the linear models, non-normalized and normalized feature training data, normalized separately from the testing data to avoid data leakage, were created, as the models will be tried on both.

For linear models, an iteration where the training and testing data were normalized. This is to investigate if removing the scale from the variables would help with model performance.

As we expect our model to be robust, the main tuning metric that we chose is the mean absolute error, instead of the mean squared error, and its root, which is more commonly used. We have chosen to do so as a model that is tuned to the mean absolute error is more robust to outliers than a model tuned to the mean squared error (Hanley et al., 2012).

3.5 Tools and Technologies

The main tool used in building the model, along with data preprocessing, is the Python programming language, with version 3.12. To run the said programming language, multiple platforms were used: Google Colab, and PyCharm (Student license), with both hosting the Jupyter notebook service.

In building the model, the scikit-learn library and the xgboost library, made for the xgboost model, were primarily used. Other libraries that were used for analysis and/or data preprocessing are pandas, polars, matplotlib, seaborn, and shap.

In the creation of the prediction system for EORI, a web-based interface was developed using HTML, CSS, and JavaScript. The site serves as the main interaction layer where users can input or use previous weather conditions, adjust risk weights, and view computed EORI results and as well as the predicted system loss and power consumption. For the back-end implementation, the Flask framework was used to bind the machine-learning model to the web interface. The trained models for both models were loaded directly into the Flask server, which processes user inputs through REST API endpoints. These endpoints receive feature values, execute the selected model, and apply min-max scaling to convert predictions into a 1–100 EORI score.

All development and application integration of the website were conducted inside Visual Studio Code (VS Code), which served as the primary IDE for file management, Python environment setup, as well as in running the system.

Chapter 4: Empirical Results and Model Analysis

4.1 Computational Environment and Experimental Implementation

• Detailed technical description of the hardware, software versions, and configuration used to run the experiments.

• Focus on reproducibility of the model training/testing, not deployment architecture.

In training the model, and in analyzing it, Python 3.12 was used, along with the following libraries: pandas 2.2.3, polars 1.33.1, numpy 2.1, matplotlib 3.10.0, seaborn 0.13.2, scikit-learn 1.6.1, xgboost 3.1.2, scipy 1.15.3, shap 0.50.0. A Windows 24H2 machine was used to run the code.

For all parameters requiring a random state seed, a default of 42 was used.

4.2 Experiment 1: Baseline Performance

As has been mentioned earlier, several candidate models were run. For the baseline model, no cross-validation was run, and is simply a one-off run. The main basis for the evaluation of model performance is the mean absolute error, prioritizing model robustness from possible shocks in the system.

The candidate models were fitted to predict system loss and power consumption, both in MWh, and separately.

System Loss

Table 4.2.1. Baseline Metrics of top 10 candidate models by MAE.

Model	MAE	MSE	RMSE	R^2	MAPE

Lasso Regression	51065.72	4.74E+09	68839.68	0.659045	0.056277
Lasso Lars Regression	51065.72	4.74E+09	68839.7	0.659045	0.056277
Linear Regression	51066.7	4.74E+09	68843.1	0.659011	0.056278
Ridge Regression	51325.1	4.63E+09	68041.74	0.666903	0.056441
Random Forest Reg	52620.16	4E+09	63231.53	0.712335	0.05784
Huber Regression Standard	57158.1	4.99E+09	70668.07	0.640693	0.06613
Gradient Boosting	57421.9	6.21E+09	78791.47	0.553339	0.063511
Lars Regression	57638.48	6.14E+09	78343.85	0.5584	0.062886
Elastic Net Regression Standard	64480.63	5.69E+09	75421.87	0.590726	0.076571
AdaBoost	66702.97	5.8E+09	76155.52	0.582725	0.076828

The linear models that were trained in non-normalized data mostly performed better in predicting system loss than the linear models that were trained on normalized data, or the ensemble models, hinting at a relatively non-complex relationship between the predictors and system loss. The top models in terms of MAE is Lasso Regression, with an MAE of 51065.72, RMSE of 68839.68, and a MAPE of 0.056277. The baseline Lasso Lars regression model exhibits the same results.metrics.

Power Consumption

Table 4.2.2. Baseline Metrics of the top 10 candidate models by MAE.

Model	MAE	MSE	RMSE	R^2	MAPE
Ridge Regression Standard	345777.2	1.73E+11	415516.1	0.789222	0.038299
XGBoost	346928.5	2.25E+11	473920.9	0.725804	0.038313
Elastic Net Regression Standard	354985.7	1.76E+11	420045.7	0.784602	0.03955

Ridge Regression	357205.5	1.92E+11	437725	0.766088	0.040041
Lasso Regression	369824.8	2.15E+11	464118.9	0.737029	0.041478
Lasso Lars Regression	369824.8	2.15E+11	464118.9	0.737029	0.041478
Linear Regression	369828.7	2.15E+11	464124.4	0.737023	0.041478
Lasso Regression Standard	385077.7	2.02E+11	449215.7	0.753646	0.042222
Lasso Lars Regression Standard	385077.9	2.02E+11	449215.8	0.753646	0.042222
Linear Regression Standard	385090.9	2.02E+11	449227.1	0.753634	0.042224

For modelling the power consumption, training the linear models in standardized data yielded the best results, with Ridge Regression having the lowest baseline MAE with 345777.2, an RMSE of 415516.1, and a MAPE of 0.038299. The ensemble model XGBoost yielded the second best, following closely with the MAE of the top baseline model.

4.3 Primary Model Results and Comparative Analysis

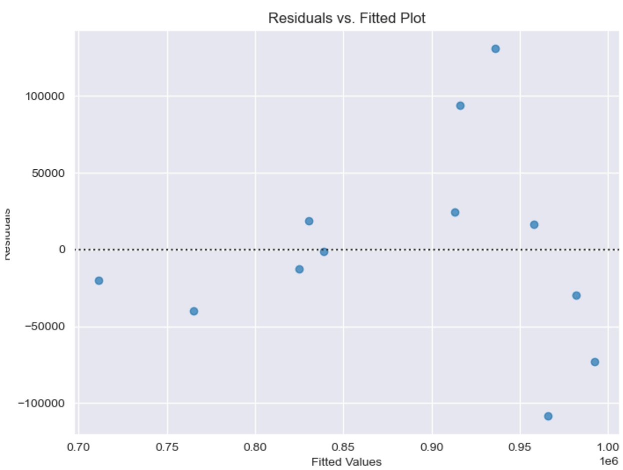
In the effort to provide a better model than the baseline, the researchers took the liberty to tune all models that reached the top 10.

System Loss

The LARS model, trained with the nonstandardized data, and with the hyperparameter n_nonzero_coeffs set to 9, provided the best metrics. This yielded a thousand MWh less in MAE than the best baseline model, and a percent reduction in MAPE. Figure 4.3.1 shows that the model's residuals are homoscedastic. Figure 4.3.2 shows that the predictions follow the actual data in a somewhat reliable manner.

Table 4.3.1. Metrics of Best Tuned Model by MAE for system loss

MAE	**MSE**	**RMSE**	**R^2**	**MAPE**
50086.65517	4119777171	64185.49	0.70359	0.055176



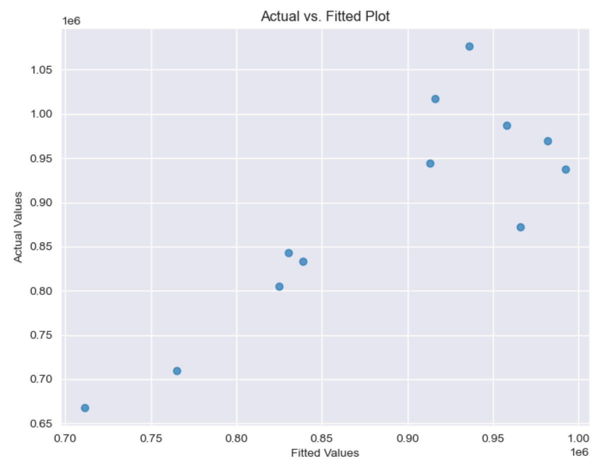


Figure 4.3.1. Residual vs Fitted plot for LARS model, and Figure 4.3.2. Actual vs Fitted plot for LARS model

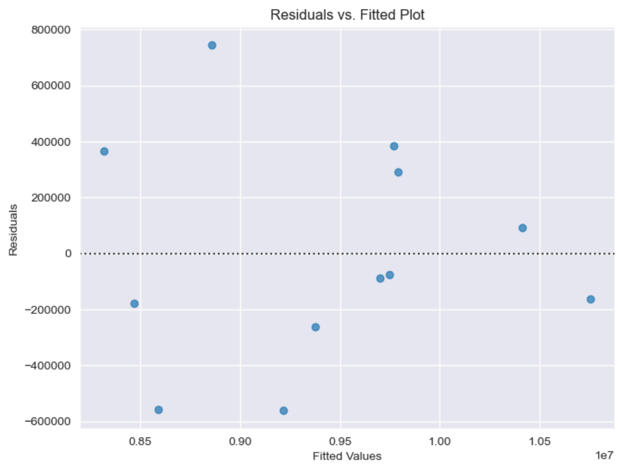
Power Consumption

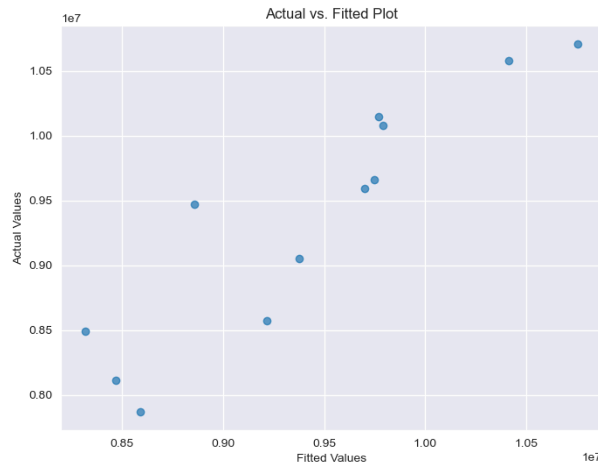
In modelling power consumption using weather hazards, the Lasso Regression model, with the hyperparameter α set to 49417.13361323838, fitted with a standardized training dataset, yielded the best results: around 20000 MWh less in MAE than the best baseline model, and a 2% reduction in MAPE.

Table 4.3.2. Metrics of Best Tuned Model by MAE for power consumption

MAE	**MSE**	**RMSE**	**R^2**	**MAPE**

324880.7	1.54E+11	391861	0.812538	0.036408





Figures 4.3.3. Residuals vs Fitted plot for Lasso Model, and Figure 4.3.4. Actual vs. Fitted plot for Lasso Model

4.4 Feature Importance and Model Explainability

With our linear models, we might see a glimpse of the contribution of each feature to the model's prediction in the coefficients. However, this rarely tells the whole story, especially when each feature might have different scales. The SHapley Additive exPlanations (SHAP) by Lundberg and Lee (2017), and its linear explainer, enables us to look not just at the coefficients, but also enables automatic standardization, and takes into account the collinearity between the features, allowing them to share the contribution.

System Loss

As can be seen from Figure 4.4.1 and Figure 4.4.2, the oceanic nino index is the most influential feature, with high and medium values influencing the estimation of system loss positively, and otherwise if negative. Low values of mean sunshine total have a great negative effect on the estimation, with the relative humidity and minimum temperature having the same effect, albeit more muted. Wind speed, the madden-julian oscillation phase, total number of typhoons, precipitation, nearby regional SST anomaly, and the max temperature, has a more muted influence on the estimate.

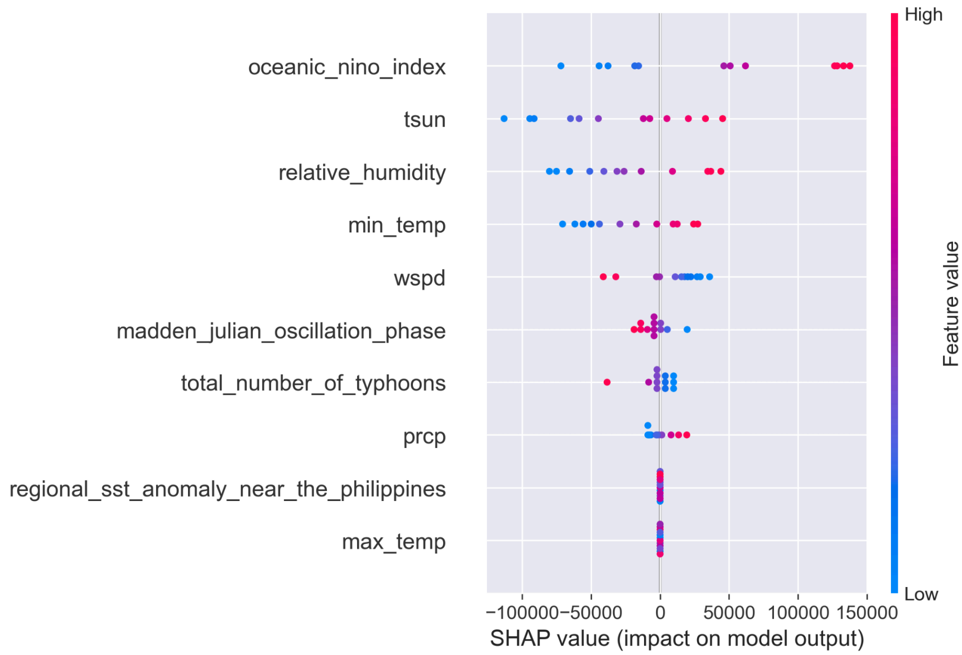


Figure 4.4.1 Beeswarm plot of SHAP values of features involved in System Loss interpolation

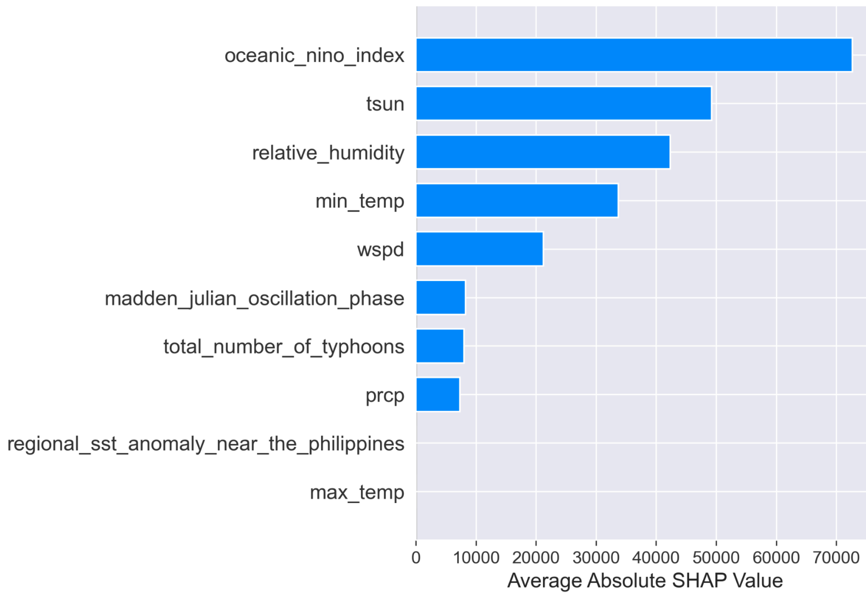


Figure 4.4.2 Bar Plot of average absolute SHAP values of features for System Loss Interpolation

Power Consumption

Figure 4.4.3 and Figure 4.4.4 show that, similar to the estimation of system loss, the most influential feature in estimating power consumption is the oceanic nino index, with high values having a huge positive influence on predicting power consumption, and low values influencing in the other direction. Minimum temperature, relative humidity, and nearby regional SST anomaly have the same effect, but in decreasing influence, with respective order. Due to the LASSO model minimizing some features to zero, only four features had SHAP values.

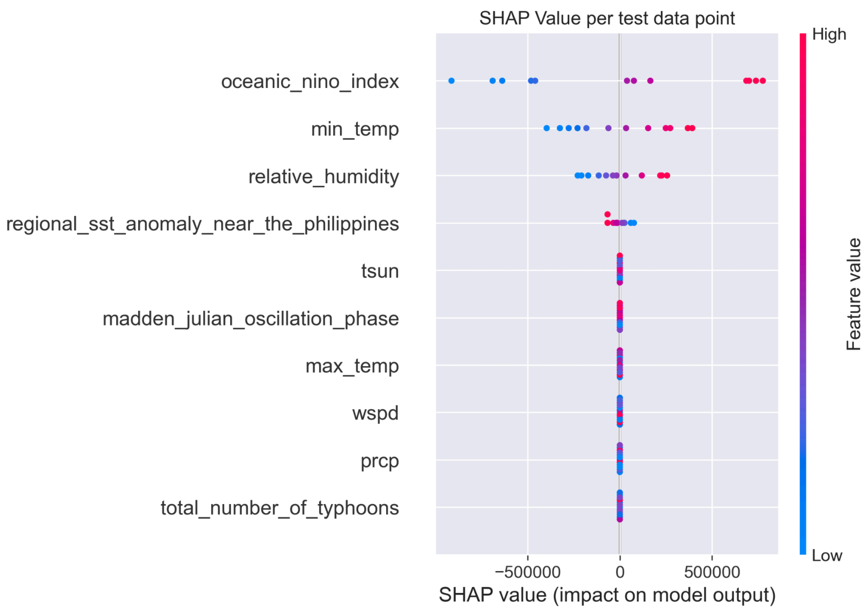


Figure 4.4.3. Beeswarm plot of SHAP values of features involved in Power Consumption interpolation

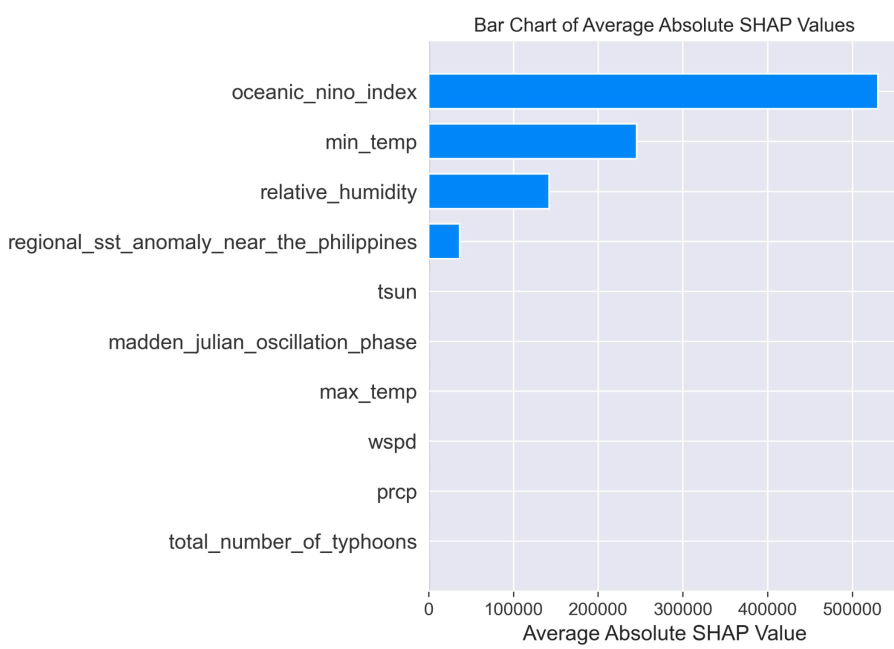


Figure 4.4.2 Bar Plot of average absolute SHAP values of features for Power Consumption Interpolation

4.5 Discussion of Results

The error values of the resulting models show that the available weather hazard data can somewhat reliably interpolate both monthly system loss and power consumption. Interestingly, it seems that the relationship is somewhat simple, given that the linear models LARS and LASSO, respectively, could explain the variance of the data satisfactorily.

Additionally, in interpolating both variables, the same features have been significantly present in estimating both monthly system loss and power consumption. Namely, the oceanic nino index, minimum temperature in a given month, and relative humidity, In addition, the average total sunshine in minutes seems to influence the system loss of the electrical grid, while the nearby regional SST affects the monthly power consumption.

Chapter 5: Conclusion and Future Work

5.1 Summary of Findings

The Philippines is still currently undergoing data infancy, despite having adopted it more than 30 years ago, and having about 87 million internet users in 2024 (Brillantes, 2025; CIA, 2024). One such area of data deficiency is in the data collection regarding public utilities, and a lack of data-driven insights regarding the subject as a result.

This study seeks to alleviate a part of this problem by providing an alternative method of enabling such insights: to create a model that can interpolate the power consumption and system loss using weather and oceanic hazard indicators.

In doing so, the researchers have built a model that can interpolate both system loss and power consumption, with an MAPE of 6% for interpolating system loss and 4% for power consumption. The model is also expected to be robust to sudden shocks in outliers in weather and oceanic hazard indicators.

As can be seen from the models, the oceanic Niño Index proved to be the most significant predictor, with the minimum temperature and the relative humidity following next.

5.2 Project Contribution and Significance

The successful outcomes of this project provide several notable contributions with direct real-world impact. Most notably, the EORI model is instrumental in Enabling Proactive Disaster Management. The model shifts electricity grid management from a reactive response to a proactive forecasting paradigm. By providing an EORI score, the model offers critical lead time, enabling utilities to strategically pre-position repair crews, stock emergency supplies, and communicate effectively with local governments. This capability can directly reduce the duration of outages and minimize socio-economic disruption following severe weather events.

A key academic contribution lies in bridging a critical data gap. The research delivers a validated methodology for quantitative risk assessment in environments characterized by data scarcity, a persistent challenge in developing regions. The EORI serves as a powerful composite risk index, effectively proving that high-utility predictive systems can be successfully constructed and validated even when ideal, comprehensive ground-truth outage data is completely unavailable.

Furthermore, the project succeeds in Promoting Policy Transparency. The integration of Explainable AI via SHAP ensures that every EORI prediction is accompanied by a quantifiable explanation of feature importance. This transparency is vital for informing policy decisions, justifying infrastructure investment in resilience measures such as typhoon-resistant facilities, and fostering trust with regulatory bodies and the public.

5.3 Limitations and Challenges Encountered

While the project was highly successful in achieving its core objectives, it is essential for academic rigor to acknowledge the inherent constraints and limitations encountered. The primary conceptual limitation lies in the Electricity Outage Risk Index's nature as a synthetic proxy for outage risk. While the index proved a necessary and effective solution to overcome the severe data scarcity, the lack of publicly accessible and consistent historical electricity outage data means the model's predictions have not yet been fully validated against a robust and standardized dataset of actual power failure events.

A significant challenge was imposed by the available electricity data's structure and frequency. The data essential for calculating the historical EORI score was available only as annual aggregates before 2020. This temporal constraint was a major limitation, as it made it impossible to conduct a more granular, month-by-month or quarter-by-quarter historical analysis for earlier years. The availability of monthly data for earlier years would have significantly increased the sample size of the time series used for training, potentially making the initial time series models much more robust and capable of capturing long-term seasonality and cyclical patterns. Furthermore, a lot of data was aggregated to a national level. This necessitated a macro-level EORI prediction and prevented the development of more personalized and targeted reports for specific provincial or regional jurisdictions, which would be crucial for localized resource allocation and reporting.

Another key challenge encountered was the Computational Intensity of Ensemble Modeling. The strategic decision to employ XGBoost within a stacked ensemble architecture was critical for achieving high predictive accuracy due to the complex and non-linear relationships in the climate data. However, this complexity introduced considerable computational overhead. As a result, the training and hyperparameter tuning of the final model were significantly more resource-intensive and time-consuming compared to developing and running simpler, more traditional linear models.

Finally, the overall model's predictive power is limited by its Lack of Micro-Grid Granularity. The EORI is a general risk indicator because the current framework does not integrate high-resolution, specific geospatial infrastructure data. This absence of granular asset data restricts the system's ability to provide targeted, micro-level maintenance recommendations at the asset level.

5.4 Recommendations for Future Work

The foundational work completed on the Electricity Outage Risk Index (EORI) model opens several avenues for future research and practical enhancement, particularly in addressing the noted limitations.

- **EORI Validation and Refinement**: The primary recommendation is to conduct a post-deployment empirical validation of the EORI against any newly released or privately acquired ground-truth outage data. Future work should focus on securing anonymized, standardized outage reports from utility partners to calculate a true, historical Outage Rate, which can then be used to validate, fine-tune, or even replace the synthetic EORI metric, transitioning the model from a proxy-based system to a direct-prediction system.
- **Geospatial and Regional Granularity**: To overcome the current limitation of macro-level prediction, the next iteration should integrate high-resolution geospatial data to create a Provincial/Regional EORI. This would involve sourcing or synthesizing provincial-level climatic and energy data, allowing for the development of separate, localized models. This refinement would provide utilities with more personalized and actionable insights for resource pre-positioning.

- **Historical Data Enhancement**: Research efforts should be directed towards collaborating with national energy and statistics bodies to retroactively acquire or estimate monthly/quarterly energy data for years prior to 2020. A richer and longer time-series dataset would significantly enhance the robustness and accuracy of the Stage 1 climatic forecasting models by enabling them to capture a broader range of seasonal and cyclical patterns.
- **Integration of Deep Learning for Complexity**: Given the computational intensity of the current XGBoost-based stacked ensemble, future work should explore the integration of computationally efficient Deep Learning architectures such as Long Short-Term Memory or Gated Recurrent Units within the cascading framework. These models are highly proficient at handling complex time-series dependencies and may offer a superior speed-accuracy trade-off suitable for real-time deployment on lower-cost hardware.
- **Operational Optimization and Decision Support**: The EORI score should be integrated into a Mixed-Integer Linear Programming or other optimization framework. This step involves translating the predicted risk score into tangible, cost-optimized maintenance and preparedness actions, such as calculating the optimal number of crews to dispatch, the necessary pre-staging of materials, and the required power generation reserve allocation, thus transforming the forecast into a fully operational decision-support tool.
- **Interactive XAI Interface**: Develop an interactive, web-based dashboard that leverages the existing SHAP explanations. This interface should allow end-users to dynamically explore why a high-risk EORI score was issued by toggling

feature importance, enhancing the model's transparency and fostering trust in its predictions for critical decision-making.

- **Incorporating Non-Weather Factors:** In scenarios where weather conditions are constant or consistent, future work should focus on considering additional factors that may influence outage risk. These could include infrastructure age, maintenance schedules, local energy consumption patterns, and socio-economic factors such as population density or emergency response capabilities. By integrating these variables, the EORI model can provide more holistic and accurate predictions, especially during periods of stable weather when other latent risk factors become more prominent.

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Appendix A: Code Snippets

````
```{python, cleaning}
print("Dataset Info:")
print(df.info())
print("Basic Statistics:")
print(df.describe())
print("\nMissing Values:")
print(df.isnull().sum())
columns_to_convert = ['power_sales', 'system_loss', 'power_consumption']
for col in columns_to_convert:
    df[col] = df[col].str.replace(',', '').astype(float)
print(df[columns_to_convert].dtypes)
print(f"\nMissing values in 'tsun': {df['tsun'].isnull().sum()}")
df['tsun'].fillna(df['tsun'].median(), inplace=True)
print(f"\nMissing values after cleaning:")
print(df.isnull().sum().sum(), "total missing values")
print(f"Cleaned dataset shape: {df.shape}")
print("Data types after cleaning:")
print(df.dtypes)
print("\nBasic statistics of cleaned numeric columns:")
print(df[['power_sales', 'system_loss', 'power_consumption', 'tsun']].describe())
```

```{python, correlation analysis}
all_potential_hazards = [
      'total_number_of_typhoons', 'prcp', 'wspd', 'max_temp', 'avg_temp', 'min_temp',
'oceanic_nino_index',
       'monthly_sst_anomaly_in_the_nino', 'regional_sst_anomaly_near_the_philippines',
'average_wind_shear',
                          'madden_julian_oscillation_phase',      'relative_humidity',
'regional_sealevel_atmospheric_pressure',
    'pres', 'tsun'
]
exposure_candidates = ['power_consumption', 'system_loss']
print(f"Total potential hazard features: {len(all_potential_hazards)}")
print("Hazard candidates:", all_potential_hazards)
print(f"Exposure candidates: {exposure_candidates}")

print("Highly correlated feature pairs (>0.8):")
high_corr_pairs = []
for i in range(len(corr_matrix.columns)):
    for j in range(i+1, len(corr_matrix.columns)):
        if abs(corr_matrix.iloc[i, j]) > 0.8:
            high_corr_pairs.append((
                corr_matrix.columns[i],
                corr_matrix.columns[j],
                corr_matrix.iloc[i, j]
            ))

if high_corr_pairs:
    for pair in high_corr_pairs:
        print(f"  {pair[0]} vs {pair[1]}: {pair[2]:.3f}")
else:
    print("  No highly correlated pairs found (>0.8)")
```

```{python, feature selection}
def intelligent_feature_selection(df, features, correlation_threshold=0.8):
    corr_matrix = df[features].corr().abs()

    importance_order = [
````
````

```
                       'total_number_of_typhoons',   'madden_julian_oscillation_phase',
'oceanic_nino_index',
             'max_temp',  'prcp',  'wspd',  'regional_sst_anomaly_near_the_philippines',
'avg_temp',
                              'min_temp',    'relative_humidity',    'pres',    'tsun',
'monthly_sst_anomaly_in_the_nino',
        'average_wind_shear', 'regional_sealevel_atmospheric_pressure'
    ]

    to_drop = set()

    high_corr_pairs = []
    for i in range(len(corr_matrix.columns)):
        for j in range(i+1, len(corr_matrix.columns)):
            if corr_matrix.iloc[i, j] > correlation_threshold:
                col1, col2 = corr_matrix.columns[i], corr_matrix.columns[j]
                high_corr_pairs.append((col1, col2, corr_matrix.iloc[i, j]))

    print("Processing highly correlated pairs:")
    for col1, col2, corr_val in high_corr_pairs:
        if col1 in to_drop or col2 in to_drop:
            continue

        for important_feature in importance_order:
            if important_feature == col1:
                to_drop.add(col2)
                print(f"  Keeping '{col1}', dropping '{col2}' (corr: {corr_val:.3f})")
                break
            elif important_feature == col2:
                to_drop.add(col1)
                print(f"  Keeping '{col2}', dropping '{col1}' (corr: {corr_val:.3f})")
                break

    to_drop = list(to_drop)
    selected_features = [f for f in features if f not in to_drop]

    print(f"\nRemoved {len(to_drop)} highly correlated features: {to_drop}")
    print(f"Selected {len(selected_features)} features: {selected_features}")

    return selected_features

selected_hazard_features  =  intelligent_feature_selection(df,  all_potential_hazards,
correlation_threshold=0.8)
```

```
```{python, model training}
# manual switch
predictor_name = "power_consumption"
# predictor_name = "system_loss"

model_data = df[selected_hazard_features+[predictor_name]]
X = model_data.drop(predictor_name, axis=1)
y = model_data[predictor_name]

RANDOM_STATE = 42
X_train,   X_test,   y_train,   y_test   =   train_test_split(X,   y,   test_size=0.2,
random_state=RANDOM_STATE)

scaler = StandardScaler()
X_train_standard = scaler.fit_transform(X_train)
X_test_standard = scaler.fit_transform(X_test)
```
```

```
# creating function for metrics

def metrics_generation( model_name, y_test, y_pred, best_params):
    mae = mean_absolute_error(y_test, y_pred)
    mse = mean_squared_error(y_test, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, y_pred)
    mape = mean_absolute_percentage_error(y_test, y_pred)


    return {
      'model': model_name,
      'mae': mae,
      'neg_mae': -mae,
      'mse': mse,
      'neg_mse': -mse,
      'rmse': rmse,
      'neg_rmse': -rmse,
      'r2': r2,
      'mape': mape,
      'best_params': str(best_params)
    }

linear_model = LinearRegression()


ridge_parameter_grid = {
    'alpha': np.logspace(-4,4,50)
}

ridge_model = GridSearchCV(Ridge(), ridge_parameter_grid, 
scoring='neg_mean_absolute_error', cv=LeaveOneOut(), n_jobs=-1)


ridge_plain = Ridge().fit(X_train_standard, y_train)

lasso_model = GridSearchCV(Lasso(), ridge_parameter_grid, 
scoring='neg_mean_absolute_error', cv=LeaveOneOut(), n_jobs=-1)

lasso_plain = Lasso().fit(X_train, y_train)


lars_parameter_grid = {
    'n_nonzero_coefs': [2, 3, 5, 7, 9, 10]
}

lars_model = GridSearchCV(Lars(), lars_parameter_grid, 
scoring='neg_mean_absolute_error', cv=LeaveOneOut(), n_jobs=-1)
lars_plain = Lars()

lars_lasso_model = GridSearchCV(LassoLars(), ridge_parameter_grid, 
scoring='neg_mean_absolute_error', cv=LeaveOneOut(), n_jobs=-1)
ll_plain = LassoLars()


huber_parameter_grid = {
    'alpha': np.logspace(-4,4,100),
    'epsilon': [1.01, 1.1, 1.35, 1.5, 2.0, 5.0, 10.0]
}

huber_model = GridSearchCV(HuberRegressor(), huber_parameter_grid, cv=LeaveOneOut(), 
scoring='neg_mean_absolute_error', n_jobs=-1)
```

```
huber_plain = HuberRegressor()

en_plain = ElasticNet().fit(X_train, y_train)

en_parameter_grid = {
    'alpha': np.logspace(-4,4,100),
    'l1_ratio': [0.1, 0.3, 0.5, 0.7, 0.9]
}
en_model    =    GridSearchCV(ElasticNet(),    en_parameter_grid,    cv=LeaveOneOut(),
scoring='neg_mean_absolute_error', n_jobs=-1)


svr_model = SVR()

ada_model = AdaBoostRegressor(random_state=RANDOM_STATE).fit(X_train, y_train)
gbr_model = GradientBoostingRegressor(random_state=RANDOM_STATE)
xgb_model = XGBRegressor(objective='reg:squarederror', random_state=RANDOM_STATE)
rf_model = RandomForestRegressor(random_state=RANDOM_STATE)
```

```{python, tuning models}
xgb_grid = {
    'learning_rate': [0.01, 0.05, 0.1],
    'max_depth': [3, 5, 7, 9],
    'subsample': [0.7, 0.9, 1.0],
    'colsample_bytree': [0.65, 0.7, 0.9, 1.0]
}

grid_xgb = GridSearchCV(
    estimator=XGBRegressor(objective='reg:squarederror', random_state=RANDOM_STATE),
    param_grid=xgb_grid,
    scoring='neg_mean_absolute_error',
    cv=LeaveOneOut(),
    n_jobs=1
)

param_dist = {
    'n_estimators': [10, 20, 30, 50, 80, 100, 200, 300, 500],
    # 'criterion': ['squared_error', 'absolute_error'],
     'max_features': ['sqrt', 'log2', 0.5, 0.8, 1], # 'sqrt' is default for regression,
or a float fraction
      'max_depth':  [5,  10,  20,  30,  None],  #  None  means  nodes  are  expanded  until  all
leaves are pure
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    #'bootstrap': [True, False]
}

grid_rf = GridSearchCV(
              estimator=RandomForestRegressor(random_state=RANDOM_STATE,    verbose=1,
criterion='absolute_error'),
    param_grid = param_dist,
    scoring='neg_mean_absolute_error',
    cv=LeaveOneOut(),
    n_jobs=-1,
    #n_iter = 50,
    verbose = 2
)

new_parameter_grid = {
    'alpha': np.logspace(4,6,50)
}
```

```
lasso_more_tune_metrics = []

# lars lasso
lars_lasso_model_moretune = GridSearchCV(LassoLars(), new_parameter_grid, scoring='neg_mean_absolute_error', cv=LeaveOneOut(), n_jobs=-1)

lars_lasso_model_moretune = GridSearchCV(Lasso(), new_parameter_grid, scoring='neg_mean_absolute_error', cv=LeaveOneOut(), n_jobs=-1)
```

Appendix B: Data Dictionary

Column Name	Description	Unit of Measurement	Type	Role in Model
year	Calendar Year of record	—	Integer	Indexing
month	Month of record	—	Integer	Indexing
power_consumption	Total energy consumed by the grid	MWh	Float	Prediction Target (Mode-Power)
system_loss	Power lost in transmission	MWh	Float	Prediction Target (Mode-Loss)
total_number_of_typhoons	Count of typhoons within the month	Count	Integer	Feature
prcp	Monthly precipitation	mm	Float	Feature
wspd	Wind speed	m/s	Float	Feature
max_temp	Maximum temperature recorded	°C	Float	Feature
min_temp	Minimum temperature recorded	°C	Float	Feature

oceanic_nino_index	Niño 3.4 anomaly	°C anomaly	Float	Feature
regional_sst_anomaly_near_the_philippines	Sea surface temperature anomaly	°C anomaly	Float	Feature
madden_julian_oscillation_phase	MJO phase (1–8)	Index	Integer	Feature
relative_humidity	Average humidity	%	Float	Feature
tsun	Sunlight hours	min/month	Float	Feature

Appendix C: Supplemental Figures/Tables